Progressive exercises

E1 · Direct application. We define on $\mathbb{N}$ the sequences (u_n) and (v_n) by $u_n = \frac{4n-1}{n+2}$ and $v_n = n^2 - 7n$. Determine their limit, justifying the transformations used.

E2 · Application and justification. We define $u_n = \sqrt{n+1} - \sqrt{n}$ for all $n \in \mathbb{N}$. Using the conjugate expression, determine the limit of (u_n).

E3 · Method to organise. We define $u_n = \frac{\cos n}{n}$ for every integer $n \geq 1$. Bound u_n, then deduce limit of (u_n).

E4 · Choice of strategy. Next (u_n) is increasing and satisfies $u_n < 5$ for all $n \in \mathbb{N}$. Justify that converges and state what can be said about its limit. Is it necessarily equal to 5? Justify.

E5 · Synthesis or proof. We define $u_n = 3 - 2(-0{,}6)^n$ for all $n \in \mathbb{N}$. Determine the limit of (u_n). Is this sequence monotonic? Justify.